Question	Answer	Marks
1(a)	force per unit mass	**B1**
1(b)(i)	$g = GM / r^2$ $= (6.67 \times 10^{-11} \times 1.0 \times 10^{13}) / (3.6 \times 10^3)^2$	**C1**
	$= 5.1 \times 10^{-5}$ N kg^{-1}	**A1**
1(b)(ii)	mass = (960 / 9.81) kg weight on comet $= (960 / 9.81) \times 5.1 \times 10^{-5}$	**C1**
	$= 5.0 \times 10^{-3}$ N	**A1**
1(c)	similarity: e.g. both attractive/pointed towards the comet e.g. same order of magnitude	**B1**
	difference: e.g. radial/non-radial e.g. same (over surface)/varies (over surface)	**B1**

© UCLES 2017